RILEY SEOW

rileyy@alumni.stanford.edu • 818-934-5088 • linkedin.com/in/rileyseow • rileyy.vercel.app

SUMMARY

Full stack web developer who works with data-heavy systems to build performant and scalable applications. Experience working in React/Typescript, D3, Node, and SQL, with a passion for the front end. Adept at integrating diverse data modalities, writing clean, efficient code, and collaborating in cross-functional teams. Studied neuroscience and computer science via an interdisciplinary B.S. in Symbolic Systems from Stanford.

SKILLS

React, Typescript, ES6 JS/HTML/SCSS, D3, Node.js, PostgreSQL, Python, Mapbox, Git, Figma

EDUCATION

Stanford University
B.S. Symbolic Systems | GPA: 4.003 / 4.3

SOFTWARE EXPERIENCE

Software Engineer (UMD Faculty Assistant) Oct. 2023 - Nov. 2025
Center for Advanced Transportation Technology Laboratory, College Park, MD

- Delivered full stack contributions across 3 codebases: querying data from servers, databases, and external APIs, and implementing efficient processing pipelines.
- Built high-performance algorithms to process and visualize live data across map layers (e.g. traffic incidents and events, nationwide cameras, dynamic signs, transit, probe speed), handling 100K+ feature points and geometries refreshed in ~15 second intervals.
- Served as a key contributor to a multi-million-dollar revamp of the RITIS map, a real-time platform used across federal and state governmental agencies.
- Mentored undergraduate and graduate student interns.

Technical Research Intern Jun. 2020 - Aug. 2020
Stanford Center for Spatial and Textual Analysis, Stanford, CA

- Built a Python web scraper to extract, clean, and categorize character data from Schmoop and LitCharts for a project exploring textual personhood in English classic novels.
- Automated data transformation and export into structured CSV format for downstream use by the word-association modeling team.
- Created scalable, reproducible data pipelines to enable subsequent NLP analysis and modeling.

Research Assistant Nov. 2019 - Mar. 2020
Stanford Parallel Distributed Processing (PDP) Lab, Stanford, CA

- Developed Javascript code for a web-based speed arithmetic task deployed on Amazon Mechanical Turk to collect behavioral psychology data at scale.
- Collaborated with a PhD researcher to begin evaluating neural network training approaches on the collected dataset, supporting cognitive modeling efforts at the intersection of neuroscience and machine learning.

Research Assistant Jun. 2019 - Aug. 2019
UCLA Blair Lab, Los Angeles, CA

- Wrote code for an open-source GUI analyzing in-vivo calcium brain imaging data collected using UCLA Miniscope technology.
- Implemented high-performance data visualization algorithms in C# to render real-time neural activity, including max image projections, cell contour overlays, and calcium tracing graphs.